

TISUL', Russian Federation

WMO: 295570

Lat: 55.7517N

Lon: 88.2976E

Elev: 211

StdP: 98.81

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.7	-34.1	-40.4	0.1	-37.5	-36.6	0.1	-33.9	13.5	-9.6	11.8	-5.7	0.2	170	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.0	19.3	27.2	18.5	25.4	17.7	20.8	26.5	19.8	25.2	18.8	23.9	2.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.0	23.6	17.7	13.0	22.6	16.7	12.2	21.6	61.4	26.5	57.8	25.4	54.5	24.0	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.1	7.9	-39.4	32.4	4.7	1.6	-42.8	33.6	-45.6	34.5	-48.2	35.5	-51.6	36.6		
(5)			-39.7	22.4	4.4	1.6	-42.8	23.5	-45.4	24.5	-47.8	25.4	-51.0	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-16.9	-13.7	-6.0	3.2	10.4	16.4	18.3	15.9	9.4	2.3	-6.8	-13.3
	DBStd	13.76	9.60	8.15	6.88	5.53	5.46	4.02	2.87	3.58	4.67	5.63	8.34	8.73
	HDD10.0	3812	836	662	497	210	65	4	0	2	67	243	504	722
	HDD18.3	6158	1094	896	754	454	252	81	37	92	270	496	754	980
	CDD10.0	776	0	0	0	7	76	196	257	186	49	6	0	0
	CDD18.3	81	0	0	0	0	4	22	35	17	2	0	0	0
	CDH23.3	902	0	0	0	1	88	284	331	171	27	0	0	0
	CDH26.7	201	0	0	0	0	22	63	75	35	6	0	0	0
Wind	WSAvg	2.8	2.8	3.0	3.4	3.3	2.9	2.0	1.6	1.7	2.4	3.2	3.3	3.4
Precipitation	PrecAvg	506	16	14	19	32	45	73	69	76	55	45	36	27
	PrecMax	627	30	32	52	65	116	142	114	139	139	87	68	52
	PrecMin	358	5	3	2	5	13	34	26	25	6	24	16	12
	PrecStd	66	6	7	11	14	22	25	24	27	28	17	14	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	3.8	10.2	21.2	29.6	30.7	31.4	30.1	27.0	19.2	10.5	3.8
		MCWB	0.6	1.1	4.6	11.4	16.2	19.2	21.0	20.2	17.0	10.4	5.9	1.0
	2%	DB	0.4	1.4	7.1	17.7	25.6	28.5	28.9	27.2	22.6	15.9	7.5	1.5
		MCWB	-1.4	-1.0	3.0	9.4	14.3	18.6	20.1	19.4	14.9	9.2	4.5	-0.4
	5%	DB	-1.5	-0.2	5.0	15.1	23.1	26.6	26.8	25.2	19.9	13.0	5.0	-0.3
		MCWB	-3.1	-2.2	1.8	8.4	13.2	17.9	19.3	18.2	13.2	8.3	2.7	-2.0
	10%	DB	-3.8	-2.0	3.2	12.3	20.4	24.6	25.0	23.1	17.2	10.4	3.1	-2.1
		MCWB	-5.2	-3.7	0.5	6.8	12.3	17.2	18.7	17.0	12.2	6.9	1.2	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	1.4	5.4	12.1	17.6	21.3	22.4	22.0	17.7	11.8	6.9	1.6
		MCDB	3.0	3.5	9.0	19.5	27.3	27.7	28.4	27.4	24.6	17.0	9.4	3.5
	2%	WB	-1.4	-0.7	3.5	10.5	15.6	20.0	21.2	20.1	15.5	10.1	4.5	-0.3
		MCDB	0.4	1.1	6.3	16.7	23.1	26.0	26.7	26.0	21.2	14.5	7.3	1.3
	5%	WB	-3.3	-2.1	2.0	8.7	14.2	18.9	20.1	18.9	14.0	8.6	2.7	-1.9
		MCDB	-1.6	-0.3	4.7	14.3	20.9	25.0	25.4	23.8	18.8	12.8	4.7	-0.2
	10%	WB	-5.2	-3.8	0.6	7.0	12.8	17.8	19.2	17.7	12.7	6.9	1.1	-3.7
		MCDB	-3.8	-2.0	3.1	12.0	19.5	23.3	24.0	22.2	16.6	10.5	2.9	-2.1
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	10.5	10.3	11.3	11.3	10.6	10.5	10.0	7.9	7.2	8.3
		MCDBR	9.2	8.8	10.9	15.5	16.2	15.6	14.4	14.1	15.4	13.0	8.5	9.2
		MCWBR	7.9	7.3	7.7	8.9	7.4	7.1	6.6	7.0	8.2	7.6	6.0	8.2
	5% WB	MCDBR	8.7	8.9	10.1	14.3	14.1	13.1	12.2	12.6	13.6	12.2	7.8	9.0
		MCWBR	7.6	7.7	7.7	8.9	7.0	6.7	6.3	6.7	7.7	7.5	5.8	8.1
Clear-Sky Solar Irradiance	taub	0.283	0.318	0.367	0.402	0.410	0.405	0.430	0.410	0.370	0.352	0.317	0.285	
	taud	2.024	1.998	1.935	2.073	2.194	2.300	2.233	2.265	2.325	2.221	2.046	1.965	
	Ebn at Noon	587	712	767	809	830	839	806	797	780	680	530	466	
	Edh at Noon	67	101	139	141	133	121	127	116	95	82	65	56	
All-Sky Solar Radiation	RadAvg	0.82	1.80	3.18	4.40	4.95	5.82	5.23	4.25	2.80	1.46	0.72	0.50	
	RadStd	0.09	0.15	0.25	0.29	0.56	0.54	0.36	0.36	0.34	0.14	0.10	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page